

A simple immersed boundary method for OpenFOAM

Example: unsteady parallel flow
uniform pressure gradient

$$\frac{\partial u}{\partial t} = -\pi_x + \frac{1}{Re} \nabla^2 u, \quad \text{the governing equation.}$$

we integrate in the Volume Ω_{ij}

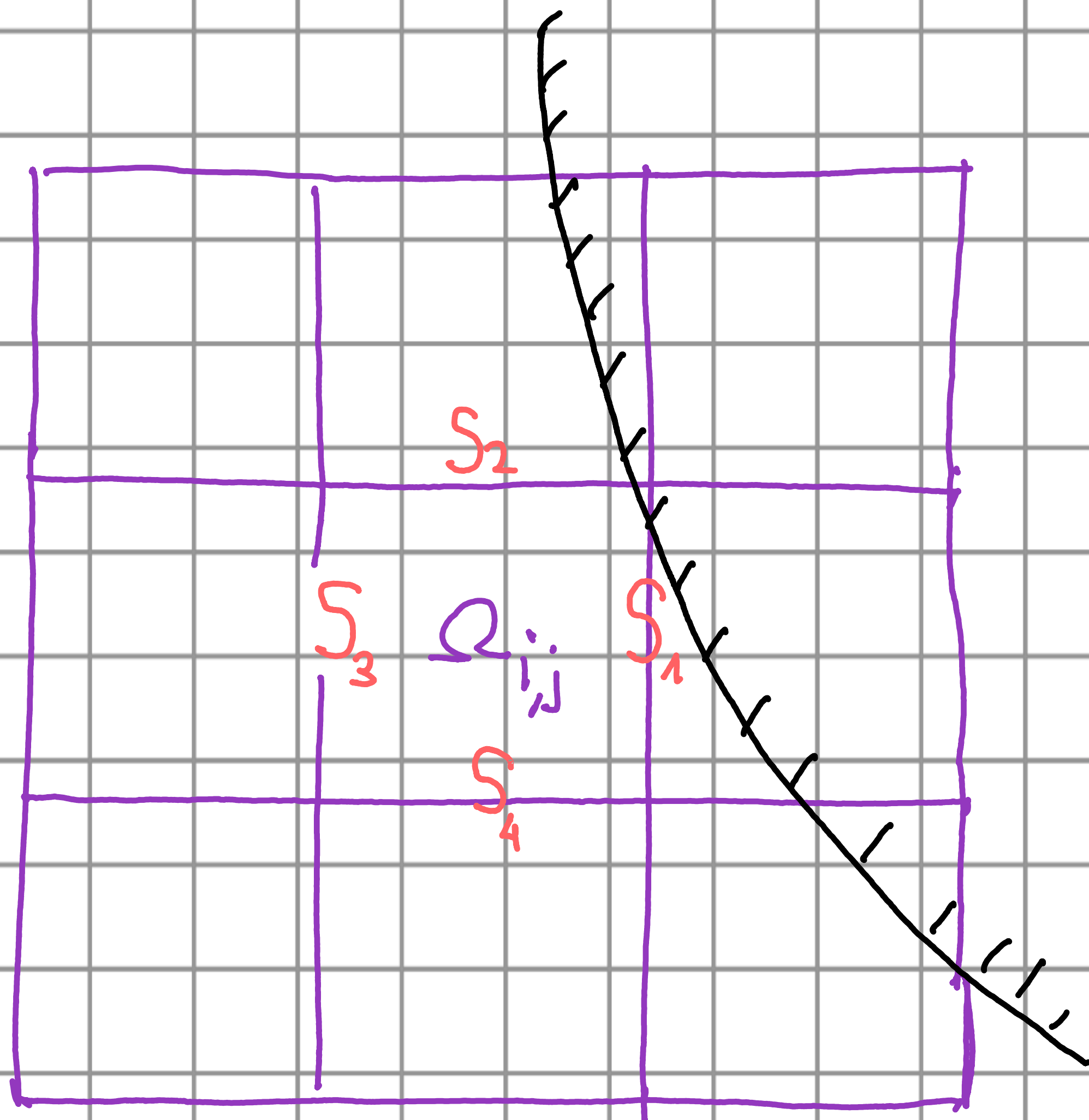
$$\frac{\partial}{\partial t} \int u d\Omega = -\pi_x \int d\Omega + \frac{1}{Re} \int \nabla^2 u d\Omega$$

substitution
of the volume
average

Gauss'
theorem

$$\frac{\partial}{\partial t} \Omega_{ij} \bar{u}_{ij} = -\pi_x \Omega_{ij} + \frac{1}{Re} \int \nabla u \cdot \mathbf{n} dS$$

THIS IS THE
ONLY TERM THAT
WE WANT TO
TREAT IN THE
IBM



$$\int \nabla u \cdot \hat{\mathbf{n}} dS = \sum_{k=1}^4 \int_{S_k} \nabla u \cdot \hat{\mathbf{n}} dS =$$

$$= \sum_{k=1}^4 S_k (\nabla u \cdot \hat{\mathbf{n}})_k$$

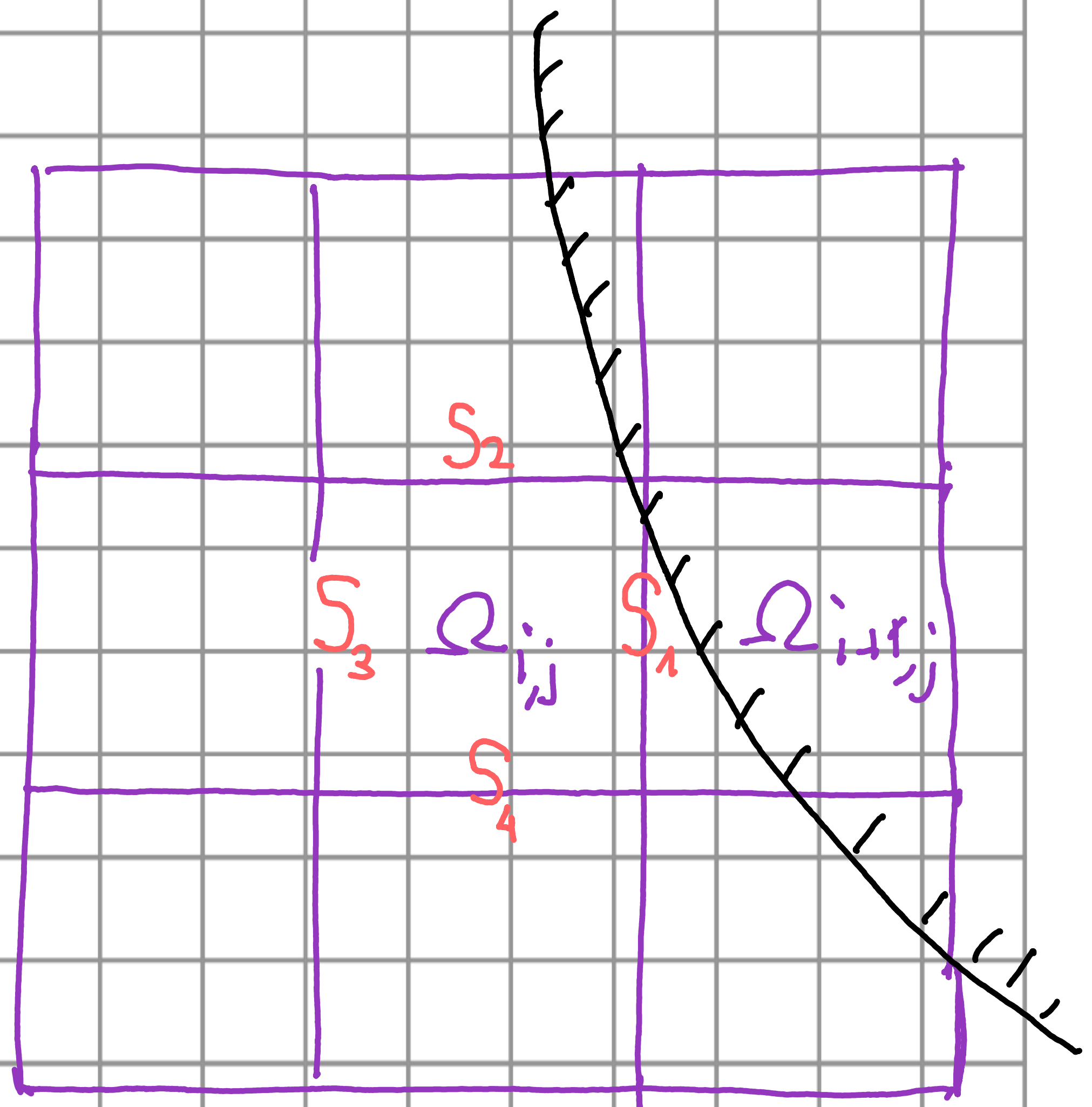
midpoint rule

let's now take as an example the surface S_1

$$S_1 \left. \frac{\partial u}{\partial x} \right|_{i+\frac{1}{2},j} = \frac{S_1 (\bar{u}_{i+1,j} - \bar{u}_{i,j})}{\Delta x}$$

2nd order
finite difference

Now since we have the body we can express $\bar{u}_{i+1,j}$ through an immersed boundary



POSSIBILITY 1: 1st-ORDER IMMERSED BOUNDARY METHOD THE "STAIRCASE" APPROXIMATION

in this case, the points in the solid are simply set to zero
this is done implicitly as follows

$$S_1 \frac{\bar{u}_{i+1,j}^{n+1} - \bar{u}_{i,j}^{n+1}}{\Delta x} - \frac{S_1 \bar{u}_{i+1,j}^{n+1}}{\Delta x}$$

STANDARD OPENFOAM OPERATOR CORRECTION

we thus must subtract a quantity from the off-diagonal entry of the system matrix

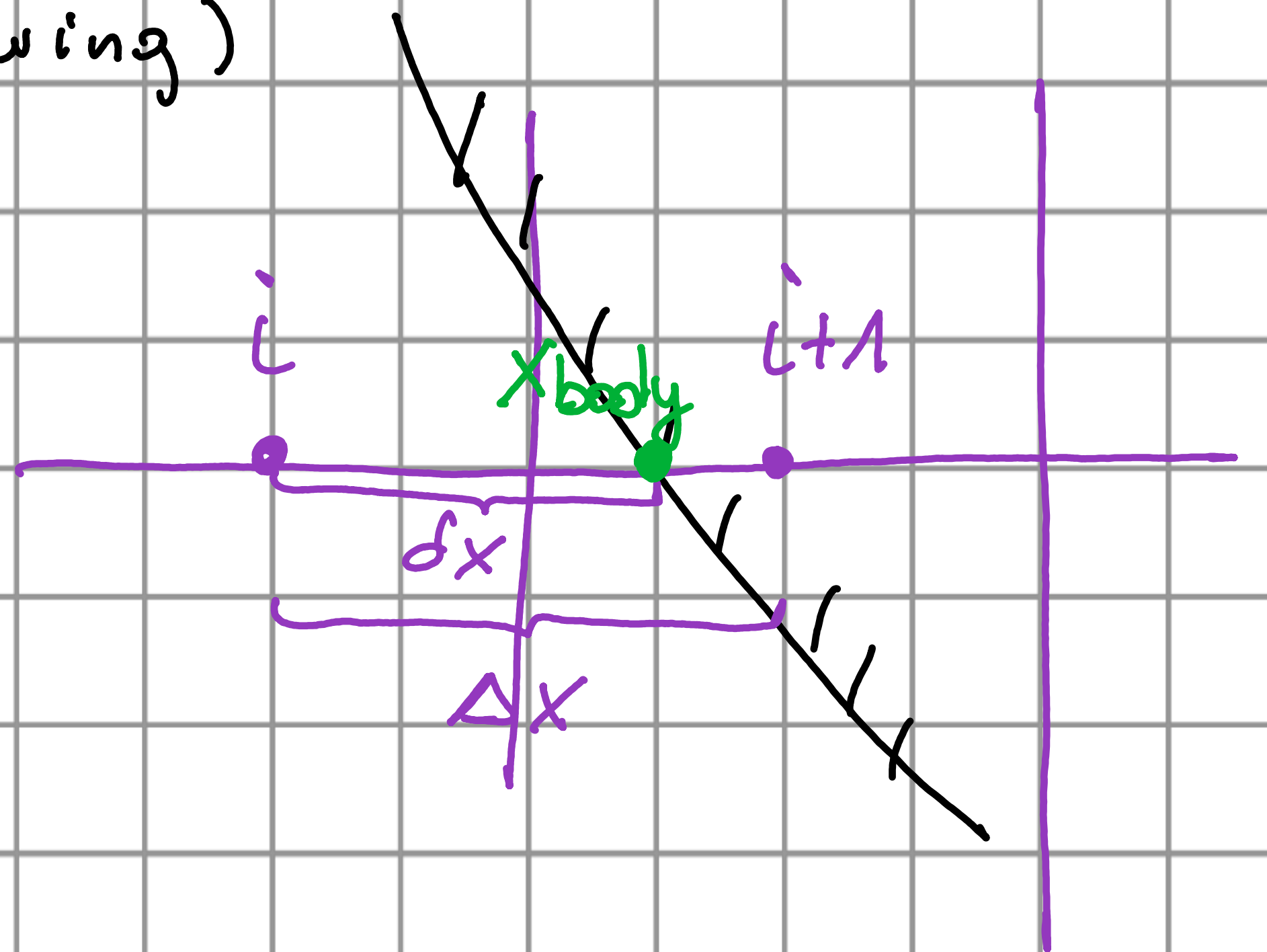
(in this case it would be sufficient to set it to zero.
However, in a general flow case there might be other contributors to that coefficient)

POSSIBILITY 2 : 2nd ORDER IMMERSED BOUNDARY METHOD

THE LINEAR EXTRAPOLATION

(I omit "j" in the following)

in this case we extrapolate linearly the value $\bar{u}_{i+1}$ using the value at (i,j) and at the body.



LINEAR EXTRAPOLATION

definitions $\Delta x = x_{i+1} - x_i$
 $\delta x = x_{\text{body}} - x_i$

$$\bar{u}_{i+1} = \bar{u}_i + \frac{\bar{u}_{\text{body}} - \bar{u}_i}{\delta x} \Delta x = \bar{u}_i \left(1 - \frac{\Delta x}{\delta x} \right)$$

typically
zero
(within body)

Again, we evaluate the Laplacian implicitly and substitute $\bar{u}_{i+1}^{n+1}$ with the linear interpolation above

$$S_1 \frac{\bar{u}_{i+1}^{n+1} - \bar{u}_i^{n+1}}{\Delta x} = S_1 \frac{\bar{u}_{i+1}^{n+1}}{\Delta x} + \frac{S_1}{\Delta x} \left(1 - \frac{\Delta x}{\delta x} \right) \bar{u}_i$$

Thus, in this case we do the same as with the staircase approximation, but we now also modify the central coefficient by adding the green term.

NOTE the modification of the off-diagonal element is not necessary IF the Laplacian is solved explicitly, only the additional IBM term implicitly, and we ensure zero velocity in the body.

However, in this way there is an additional viscous stability constraint: the grid-based Peclet number must be smaller than 0.5.

This results in very small timesteps for very low Reynolds numbers.